Networks: capacities

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Solution of the practice quiz

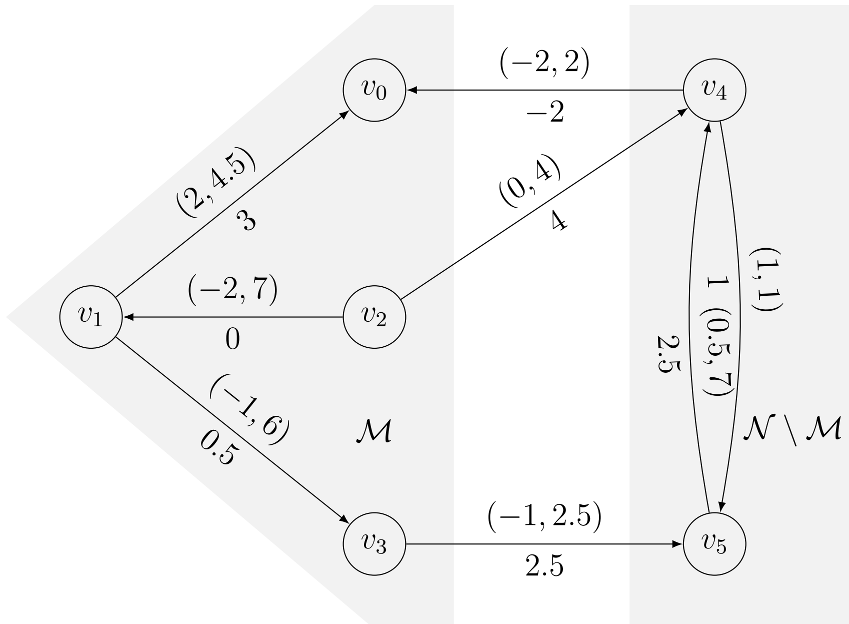
1. First, we write the cut and the sets of forward and backward arcs:

   - $\Gamma = (\mathcal{M}, \mathcal{N} \backslash \mathcal{M})$ with $\mathcal{M} = \{v_0, v_1, v_2, v_3\}$ and $\mathcal{N} \backslash \mathcal{M} = \{v_4, v_5\}$.
   - $\Gamma^{\rightarrow} = \{(v_2, v_4), (v_3, v_5)\}$ is the set of forward arcs.
   - $\Gamma^{\leftarrow} = \{(v_4, v_0)\}$ is the set of backward arcs.

2. The capacity of the cut Γ is

$$\begin{aligned} \mathcal{U}(\Gamma) &= \sum_{(i,j)\in\Gamma^{\rightarrow}} u_{ij} - \sum_{(i,j)\in\Gamma^{\leftarrow}} \ell_{ij} \\ &= (u_{24} + u_{35}) - (\ell_{40}) \\ &= (4 + 2.5) - (-2) \\ &= 8.5. \end{aligned}$$

3. To saturate the cut, the flow on each forward arc must be equal to the upper bound, and the flow on each backward arc must be equal to the lower bound. Therefore, the flow on arc (v_4, v_0) must be -2, the flow on arc (v_2, v_4) must be 4, and the flow on arc (v_3, v_5) must be 2.5. The flows on all the other arcs of the graph are irrelevant for the saturation of the cut. It must be verified that the capacity constraints are verified, though. An example of flow vector is represented in the figure below.



v_0
v_1
v_2
v_3
v_4
v_5
$(-2, 2)$
-2
$(2, 4.5)$
3
$(0, 4)$
4
$(-2, 7)$
0
$(-1, 6)$
0.5
$(-1, 2.5)$
2.5
$(1, 1)$
1
$(0.5, 7)$
2.5
$\mathcal{M}$
$\mathcal{N} \setminus \mathcal{M}$